

Applying survival analysis and Explainable Artificial Intelligence to understand academic success in software engineering students

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ABSTRACT

This paper presents a characterization of software engineering students based on their admission profiles and academic trajectories. Using a larger dataset from the 2014–2019 cohorts, the study combines statistical methods, supervised learning techniques, Explainable Artificial Intelligence (XAI), and Survival Analysis (SA) to understand and predict student outcomes. The results confirm a strong relationship between admission variables, such as age, CENEVAL score, GPA, credit progress, and intended highest educational level, and academic success or dropout risk. Multinomial Logistic Regression, Decision Trees, and Cox Proportional Hazards models were employed to classify students into three academic trajectory (AT) types: timely graduation (Type A), delayed graduation (Type B), and dropout (Type C). Multinomial Regression achieved the highest accuracy at 90%, outperforming the other models by approximately 10%. Notably, while a high CENEVAL score often indicated successful graduation, exceptions were observed, emphasizing the need for more nuanced models. Survival Analysis provided a dynamic view of academic trajectories over time, revealing that student performance can shift due to various internal and external factors. These findings offer valuable insights for educational authorities, enabling targeted interventions to reduce dropout rates and improve program retention. Future work will explore additional XAI techniques, expand the student sample size, incorporate new variables like student motivation, and perform comparative studies across different Mexican regions to evaluate geographic influences on software engineering student profiles.

1. Introduction

Software Engineering (SE) is a discipline that provides processes, tools and methodologies for the development of quality software [1,2]. Thus, SE education plays an essential role in training students on technologies, processes, and practices that ensure quality in software development [3]. However, the number of software developers with SE degrees is small compared to the number of people who develop or modify software [4].

Some studies report dropout rates of up to 40% in degree programs related to computing and technology [5]. This need for professionals in SE and related fields has increased the number of studies on school dropout in these fields [6]. The study of dropout rates in SE and related computer science fields has received increased attention due to the need for professionals in these fields [6].

For the above, our study focuses on software engineering students, as each admission profile presents distinct characteristics. Specifically, students in SE programs are expected to develop strong problem-solving and computational thinking skills, which can already be observed in

their admission test performance. Moreover, the dropout rate in this program is considerably high, making it essential to identify the factors associated with persistence and academic success. At the same time, the profile of SE graduates is highly demanded in the labor market, which further underscores the importance of supporting students throughout their academic trajectory.

Due to the above, the need arises for educational institutions to apply or adopt techniques to detect students at greater risk of abandoning their studies to design and implement effective measures to reduce the school dropout. Nowadays, it is possible to access data on the entry profile. With this information it is possible to identify characteristics of the students who enter an educational program and with Artificial Intelligence algorithms, to generate models applied to education [7–10].

School authorities ask themselves several questions to make decisions and improve the terminal efficiency of their educational programs, such as “What identifies students with good and poor academic achievement? Which are the causes of the students’ retention in the

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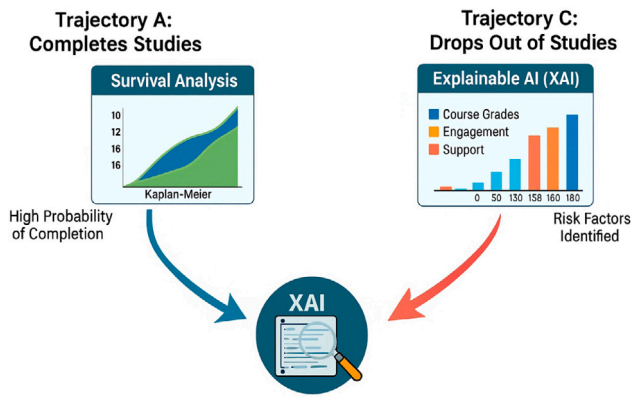


Fig. 1. Overview of the benefits of SA and XAI in our study.

university? Why do students drop out” [11]. In [12] was studied that the success of an academic trajectory (AT) in the area of software engineering is influenced by the entry profile that students present when they begin their studies.

The entry profile is defined as “the set of knowledge, skills, attitudes, and values that determine how well-prepared the applicant have” [13]. By studying these variables, the results can reveal the characteristics that students need to demonstrate, as this, in some cases, guarantees a solid professional education. On the other hand, the academic trajectory is the path a student takes within an educational system.

Analysis of the school trajectory based on the entry profile is important because the students of each region (even universities in the same country) have different characteristics that may not be generalized. Predicting the successful completion of a SE student’s bachelor’s degree can be done based on historical data and with different machine learning (ML) algorithms, such as artificial neural networks, K-nearest neighbors, naive Bayes, among others.

However, many algorithms tend to be “black boxes”, hindering their adoption in environments where transparency and explainability are essential [14,15]. This lack of interpretability limits the identification of the most influential variables in the predictions [16], which is important to detect the variables that influence students to complete or not their studies and generate strategies to avoid dropping out. In response to this problem, explainable Artificial Intelligence (XAI) emerges as an innovative solution by allowing developers to understand how predictions are generated and what factors influence them [17].

On the other hand, the application of ML could predict a student’s final academic performance based on their entry profile. However, student performance could change throughout the academic career due to various factors, including some subjects that some students may find difficult or dislike. Even extracurricular issues such as family, illness, or the need to work can influence performance over time, generating a dynamic rather than static prediction. This dynamic analysis can be studied by means of a survival analysis (SA).

Fig. 1 shows an overview of our study, where, through XAI, we not only identify the probability of completing a successful Academic Trajectories (AT) or not over time with SA, but we also identify risk factors that influence the completion of undergraduate studies in SE. This with the intention of having elements to make decisions by educational authorities.

This study contributes to the field of Computer Science by proposing and validating a transparent, interpretable, and reproducible methodological pipeline that combines SA and XAI for human-centered data analysis, an approach applicable beyond educational contexts. So, given the studies presented in Section 2 Related work, our contributions are listed below:

- We conducted and analyzed a case study on students enrolled in the Bachelor’s degree in Software Engineering at a Mexican university, given that the skills required for this discipline are highly specific, such as proficiency in mathematics, logical thinking, and even certain social traits.
- We analyzed academic trajectories using survival analysis, which not only define trajectory types but also predicts how long it will take a student to complete their studies or to drop out.
- We applied Explainable Artificial Intelligence (XAI) models to characterize students with a successful academic trajectory (AT Type A), a regular academic trajectory (AT Type B), or dropout (AT Type C), based on variables from their entry profile.
- We compared the results with a multinomial logistic regression.
- The results of the models and predictions are statistically analyzed to strengthen our conclusions.

This paper is organized as follows. Section 2 presents some studies related to our research. Section 3 describes the method used to relate the entry profile and the academic trajectory. Section 4 presents the results, analysis and findings. Section 5 discusses the threats to the validity of the study. Section 6 draws the conclusions. Finally, Section 7 presents the future work.

2. Related work

We divided the related studies into three subsections. The first presents some works that have addressed the topic of academic trajectories. The second summarizes recent studies that have used SA and XAI methods to predict school dropout rates. Finally, we group works focused on software engineering and student analytics.

2.1. Prediction of academic trajectories

Several studies have highlighted the importance of analyzing student data to prevent potential dropout. In [11], the authors analyzed a large academic database of the Industrial University of Santander (20,000 students over fifty years) using the C4.5 algorithm, generating 331 decision rules. The strength of this study lies in the scale of the dataset and the ability to derive diverse rules for performance and dropout. However, a limitation is that entry profiles vary significantly across disciplines, making generalization across fields of knowledge difficult.

The study in [18] examined 11,466 admissions at the Universidad Nacional del Centro de Perú using regression and decision tree models, achieving 95.44

In [19], a tri-branch CNN with attention mechanisms was proposed to capture persistence and temporal behavior. This approach is innovative, leveraging deep learning to detect complex patterns. Its limitation, however, is the computational cost and the need for large labeled datasets, which may restrict its applicability in resource-constrained educational contexts.

An early survival analysis (SA) approach is found in [20], which used the Cox regression model to study how combinations of career and technical education (CTE) and academic courses influence dropout. A strength of this work is the use of a robust statistical method to show that CTE participation increases motivation and attachment. However, its limitation is that it focuses only on high school contexts, reducing its transferability to higher education.

In [21], a Cox-based SA framework was applied at Wayne State University, incorporating time-varying factors and pre-enrollment data. Its strength is the use of early predictors such as credits passed/failed and GPA as actionable early-warning indicators. A limitation is that the results are institution-specific, which may affect generalizability.

Finally, [22] used a Log-Normal regression model to identify factors influencing dropout among Computer Science students at the University of Brasilia. The advantage of this study is the focus on both academic and social performance factors, offering a holistic view. However, as with other single-institution studies, the findings may not hold in different cultural or institutional contexts.

Table 1
Related studies applying SA and XAI to address school dropouts at university level.

Study	Year	Source	Model
[21]	2016	ACM	Cox proportional hazards model
[23]	2018	ACM	Cox proportional hazards model
[24]	2019	Elsevier	Epanechnikov Kernel-Smoothed Hazard Function
[25]	2025	Elsevier	SHAP and LIME
[12]	2024	IEEE	Logistic regression model
[22]	2023	IEEE	Kaplan–Meier model, Log-Normal regression model
[26]	2021	IEEE	Cox proportional hazards model, Kaplan–Meier model
[27]	2021	IEEE	Cox proportional hazards model, Kaplan–Meier model
[28]	2014	IEEE	Cox proportional hazards model
[29]	2022	Springer	Kaplan–Meier model
[30]	2018	Springer	Cox proportional hazards model
[31]	2018	Springer	Exponential modeling
[32]	2015	Springer	Logistic regression
[33]	2013	Springer	Multinomial logistic regression model
[34]	2016	Wiley	Cox proportional hazards model
[35]	2016	Wiley	Multinomial logistic regression model
[36]	2015	Wiley	Logistic regression
[37]	2024	Taylor & Francis	Kaplan–Meier model
[38]	2024	Taylor & Francis	Kaplan–Meier model
[39]	2022	Taylor & Francis	Cox proportional hazards model
[40]	2017	Taylor & Francis	Multinomial logistic regression model
[41]	2019	Taylor & Francis	Kaplan–Meier model
[42]	2024	Taylor & Francis	Cox proportional hazards model

2.2. Survival analysis and explainable artificial intelligence in university dropout rates

Table 1 presents the studies found in different data sources on SA and XAI at the university level.

Table 1 highlights that the Cox proportional hazards model is among the most frequently used approaches. Introduced by D. R. Cox in 1972, this model was designed to estimate the impact of various covariates on system failure times [43]. Its frequent application establishes Cox as the most commonly used SA technique for identifying students at risk of dropping out. Studies such as [21,39] employed Cox models for early dropout risk detection. Particularly, works by [21,26,30], and [34] used the Cox model to examine variables influencing dropout, successfully identifying key risk factors. Its strength lies in providing interpretable risk estimates, yet its reliance on proportional hazard

assumptions can reduce flexibility when dropout patterns deviate from this structure.

Variations of the Cox model have also emerged, such as the Shared Frailty Cox model proposed in [38], which analyzes the academic careers of students across different engineering programs. This model stands out as a relatively recent approach in dropout prediction research, offering the advantage of accounting for students' career progression and modeling time-dependent variables.

The Kaplan–Meier estimator ranks as the second most frequently utilized technique among the primary studies reviewed. Its nonparametric nature—requiring no assumptions about the time-to-event distribution [44], which makes it particularly suitable for contexts where the temporal characteristics are unknown. Its flexibility contributes to its widespread use in school dropout studies. This model has been applied across various studies [37,39,41]. In particular, [37] focused on identifying pre-enrollment factors linked to student retention in Sport, Health, and Exercise Sciences programs in the United Kingdom. The findings revealed that male students, those from neighborhoods with low participation in higher education, and students entering without A-Level qualifications faced a greater risk of failing to complete their degrees. However, this approach lacks explanatory power for identifying specific covariates influencing dropout.

Regarding the use of Explainable Artificial Intelligence (XAI), in [32] was implemented a logistic regression model to estimate the probability of students changing majors. Similarly, [23] applied this method to link event risks to predictive variables. Additionally, [36] used logistic regression to assess the effect of employment on university students. A comparable method, multinomial logistic regression, was employed by [33] to analyze the relationship between event risk and various covariates. This approach was further expanded upon by [35,40], who used it to evaluate the influence of different factors on academic outcomes. The advantage of XAI lies in transparency and low computational demand, though they often underperform in capturing time-to-event dynamics compared to SA models.

The most closely related study to ours is the one reported in [25], where the authors employed XAI, specifically SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-Agnostic Explanations), to predict undergraduate students' academic performance in the context of Bangladesh. However, while SHAP and LIME have become widely used in XAI research, both methods present important limitations. SHAP is computationally expensive, especially for complex models and large datasets, which can hinder scalability. Moreover, the explanations it produces may be difficult to interpret for non-technical stakeholders, as they often require advanced statistical knowledge. LIME, on the other hand, relies on generating local approximations of the model, which can result in unstable or inconsistent explanations depending on the sampling strategy and the perturbations introduced. These drawbacks highlight the need for complementary or alternative approaches to ensure robust, interpretable, and scalable explanations in educational prediction tasks.

However, no studies were found in Mexico specifically addressing the academic trajectory of Software Engineering students based on their admission profiles. Nevertheless, existing research reflects a national interest in studying undergraduate student success.

For instance, [45] developed a decision tree model to predict dropout among higher education students from private institutions, using 30 variables, including previous academic performance, age, sex, number of credits passed and failed, among others. The findings identified previous academic performance, current academic performance, age, and gender as some of the most influential factors.

Another study, [46], explored the relationship between admission profiles and academic performance among students from the Psychology Academic Unit from the 2009–2014 cohort. Focusing initially on four personality dimensions (emotional stability, openness, responsibility, and professional success). The study involved a sample of 106

students, mostly women. Using a quantitative, correlational, cross-sectional cohort design, and applying the TEA Personality Test alongside a chi-squared analysis, the authors concluded that no significant relationship existed between admission profiles and either academic performance or personality traits.

2.3. Analysis of software engineering students

In the field of Software Engineering, various studies have focused on students for different objectives, such as evaluating Software Engineering curricula [47–49], applying expert systems [50], using games and gamification strategies in Software Engineering [51], exploring the relationship between academia and the software industry [52–55] [56], and proposing strategies and tools for specific courses within the Software Engineering curriculum [57,58], among others.

In conclusion, while there are numerous studies across different educational contexts aimed at reducing dropout rates and identifying students at risk of not successfully completing their academic programs, and despite research efforts focused on evaluating curricula and motivating students in different SE courses, no study has been found that assesses applicants' entry profiles to predict their academic trajectory. Additionally, it is important to note that, in Mexico, there are no studies specifically addressing the academic trajectory within Software Engineering programs, considering that the entry profile for this field can differ significantly from that of other disciplines.

3. Research method

This study concentrated on identifying the shared variables and methodologies employed by the authors referenced, aiming to explore the connection between students' entry profiles and their academic trajectories. An analysis of the reviewed literature revealed that studies using similar variables often applied the same statistical techniques and modeling approaches to determine the key characteristics influencing student performance. These methods were primarily used to forecast academic success or failure and to pinpoint the factors associated with student dropout.

3.1. Data collection

The data utilized in this research were obtained from institutional systems that store applicants' admission profiles. These profiles include personal details such as gender, age, and place of origin, as well as the applicants' academic background, particularly those who took the National Higher Education Entrance Examination (EXANI-II). This exam evaluates students' aptitude and knowledge across four domains: Mathematical Thinking, Logical-Mathematical Reasoning, Language Structure, and Reading Comprehension.

In addition, academic performance indicators such as the grade point average (GPA) and credit completion progress at the time of completion of the degree were considered. The study population comprised students admitted to the Software Engineering program in the years 2014 through 2019, corresponding to cohorts expected to graduate between 2018 and 2025.

3.2. Data preprocessing

When an applicant submits their admission application, information about their previous studies and socioeconomic background is collected. However, this instrument is updated each year, and the questions it contains may vary from one year to the next. Therefore, only the questions (variables) that appeared in the six admission years studied were retained, while variables do not present in every year were eliminated.

We classified students, based on their academic progress, into the following three types of Academic Trajectory (AT).

- AT type A: Successful academic trajectory, completed in 8 or 9 semesters.
- AT type B: Regular academic trajectory; the student failed some subjects but ultimately completed their studies.
- AT type C: The students did not complete their studies.

3.3. Data description

The dataset includes information on 421 students and consists of 33 variables (20 qualitative and 13 quantitative).

Table 2 provides a detailed overview of each variable retrieved from the institutional platform. These variables are organized into four main categories: general characteristics, socioeconomic background, prior education, and section scores from the EXANI-II assessment.

For the purpose of analysis, additional variables related to academic trajectories were incorporated. These were classified as A, B, or C: Category A includes students who completed their degree within the expected time frame, Category B includes those who finished with some academic challenges, and Category C refers to students who discontinued their studies. Additionally, a time-related variable was added, which captures the enrollment date, the exit date (indicating either dropout or continued enrollment), and a survival status indicating whether the student left the program or remains active.

3.4. Descriptive analysis

A descriptive and exploratory analysis was carried out to offer a comprehensive overview of the population's characteristics, focusing on its distribution and proportions to identify potential visual patterns. This analysis primarily facilitated the generation of frequency tables, individual variable plots, and data distribution visualizations, all of which supported the subsequent application of appropriate correlation analyses.

The number of students enrolled during the study period was recorded. As shown in Table 3, enrollment figures were as follows: 57 students in 2014, 73 in 2015, 76 in 2016, 67 in 2017, 70 in 2018 and 78 in 2019.

Upon examining the variables, it was observed that male students outnumbered female students throughout the analyzed period. Table 4 presents this breakdown, indicating that 54 women were enrolled—of whom 19 had a successful academic trajectory (AT type A), 10 had a regular trajectory (AT type B), and 25 dropped out (AT type C). In comparison, 367 men were enrolled, with 97 achieving a successful trajectory, 48 following a regular path, and 222 dropping out.

Fig. 2 shows the distribution of SE students by gender who experience the event in question for survival analysis: dropout. This plot reveals that the majority of students do not complete their bachelor's degree in Software Engineering at the examined Mexican university, highlighting the need to investigate the factors contributing to student dropout.

The CENEVAL score represents the result students achieve on the EXANI-II exam. As shown in Fig. 3, the distribution follows a normal pattern, centered on 75 out of 100 points. This indicates relatively high scores compared to the national average of all applicants seeking admission to higher education.

3.5. Correlation between variables

To determine which variables from the Admission Profile are associated with students' academic trajectories, two statistical methods were employed: Pearson's chi-squared test and Spearman's rank correlation coefficient. Pearson's chi-squared test is suitable for analyzing the relationship between two categorical or qualitative variables. In this study, the dependent variable is academic trajectory, which is classified into three categories (TA Type A, AT Type B, and AT Type C). Accordingly, this test was appropriate for exploring associations

Table 2
Variables in the dataset.

Variable	Response	Type
Year	Since 2014 to 2019	Continuous
Academic Trajectory	A, B or C	Ordinal
Ceneval score	0 to 100 points	Continuous
Logical-Mathematical thinking	0 to 100 points	Continuous
Mathematical Thinking	0 to 100 points	Continuous
Language Structure	0 to 100 points	Continuous
Verbal Thinking	0 to 100 points	Continuous
Age	17 to 40 years	Continuous
Gender	Male, Female	Discrete
State of Birth	State where the program is offered, Other	Discrete
Mother's Dialect	Yes, No	Discrete
Father's Dialect	Yes, No	Discrete
School Regime	Private, Public	Discrete
Certificate Modality	General High School, Technological High School	Discrete
School Shift	Morning, Mixed, Afternoon, Other	Discrete
Scholarship for Economic Need	Yes, No	Discrete
Level of Education to be Achieved	Bachelor, Postgraduate	Discrete
Mother's Education Level	Without Studies, Elementary School, High School, Bachelor, Postgraduate	Discrete
Father's Education Level	Without Studies, Elementary School, High School, Bachelor, Postgraduate	Discrete
Washing Machine at Home	Yes, No	Discrete
Refrigerator at Home	Yes, No	Discrete
Microwave at Home	Yes, No	Discrete
Internet at Home	Yes, No	Discrete
Pay TV at Home	Yes, No	Discrete
Computers at Home	0 to 4	Discrete
Televisions at Home	0 to 4	Discrete
Cars at Home	0 to 4	Discrete
Bathrooms at home	0 to 4	Discrete
Grade Point Average (GPA)	0 to 10	Continuous
Credit Progress	0 to 110 credits	Continuous
Survival	0 = desert/1 = stay	Nominal
Start Date	August 2019 to august 2022	Nominal
End Date	January 2020 to January 2024	Nominal

Table 3
Number of students classified by year of entry into the software engineering program.

Year	A	B	C	Total
2014	11	11	35	57
2015	20	17	36	73
2016	7	8	61	76
2017	21	8	38	67
2018	33	2	35	70
2019	24	12	42	78
Total	116	58	247	421

Table 4
Gender of students by AT.

Gender	A	B	C
Female	19	10	25
Male	97	48	222
Total	116	58	247

between academic trajectory and other qualitative variables, such as gender and the highest level of education the student intends to pursue.

In contrast, Spearman's rank correlation is used to examine relationships between a qualitative variable (after coding) and a quantitative variable. This method was applied to assess which components of the EXANI-II exam (such as the overall CENEVAL score, Mathematical Thinking, Logical-Mathematical Reasoning, Language Structure, and Reading Comprehension) were correlated with students' academic trajectories.

To select the relevant variables, p-values were analyzed, and based on the corresponding hypotheses, the following variables were identified:

- H0: There is not a relationship between academic trajectory and the variables of entry profile.

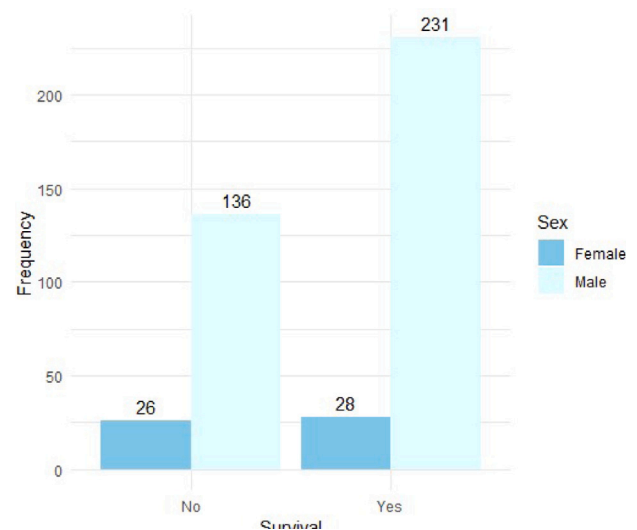


Fig. 2. Distribution of SE students by gender and survival.

- H1: Exist relationship between academic trajectory and the variables.

A total of nine variables from the admission profile were found to be significantly associated with students' academic trajectories at the AT. Table 5 presents the results of the statistical tests, all demonstrating significance at the 95% confidence level.

Before conducting the multinomial regression, it was necessary to ensure that the included variables were not highly correlated, and that the assumption of homoscedasticity was satisfied. As shown in Table 6, the Variance Inflation Factor (VIF) values for all variables

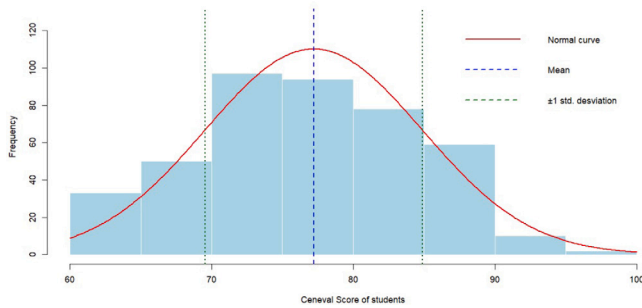


Fig. 3. Distribution of CENEVAL scores, with mean and standard deviation indicated.

Table 5
Significant variables.

Variable	P-value
Ceneval score	0.0294
Washi machine at home	0.0352
Mathematical thinking	0.0000
Language Structure	0.0000
Verbal Thinking	0.0000
Age	0.0032
Mother's Education Level	0.0002
Grade Point Average (GPA)	0.0000
Credit Progress	0.0000

Table 6
Variance Inflation Factor (VIF).

Variable	VIF
Ceneval score	2.9017
Washi machine at home	1.0670
Mathematical thinking	1.8903
Language Structure	1.6235
Verbal Thinking	1.6085
Age	1.0404
Mother's Education Level	1.0178
Grade Point Average (GPA)	1.1824
Credit Progress	1.1979
Credit Progress	0.0000

were below 5, confirming the absence of multicollinearity. To verify homoscedasticity, a plot of standardized residuals versus fitted values was generated. Fig. 4 displays a random scatter without discernible patterns, which supports the assumption of homoscedasticity.

3.6. Explainable artificial intelligence

Artificial Intelligence is the area of Machine Learning (ML), which is defined as a field of study that allows computers to learn from experience [59]. On the other hand, Goodfellow et al. [60] describe it as a set of techniques that allow systems to identify patterns in data and make decisions based on them. These perspectives show the importance of data in the learning process. However, a persistent challenge in the use of these ML algorithms is the lack of interpretability of the models, known as black box models, which make it difficult to understand how and why they make certain decisions [16]. In this context, Explainable Artificial Intelligence (XAI) emerges.

Authors such as Molnar [16] define XAI as a set of techniques that allow users to understand how AI models make decisions, identifying which variables influence predictions and why certain decisions were made. Complementing this definition, Ribeiro et al. [17] point out that

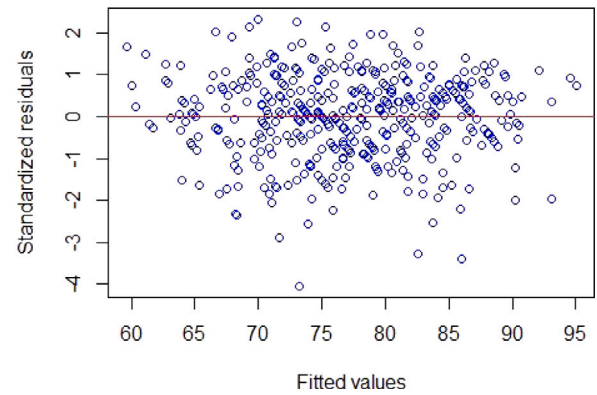


Fig. 4. Plot of standardized residuals versus fitted values.

XAI not only improves the interpretability of models, but also increases user confidence in the results.

In this study, we applied two XAI models to help identify the variables that most influence the success or dropout of an AT: Multinomial Logistic Regression and Decision Trees.

Multinomial logistic regression extends binary logistic regression by accommodating dependent variables with more than two categories. Since our dependent variable consists of three categories (AT type A, AT type B, and AT type C), this method was appropriate for classifying SE students and their AT. In multinomial regression, it is necessary to define a reference category from which the other categories are predicted, aiding the interpretation of the regression coefficients.

Before building the multinomial logistic regression model (MLRM), the assumptions of homoscedasticity and absence of multicollinearity were verified as shown in Section 3.5.

Decision trees are a supervised machine learning method commonly used for classification and decision-making tasks. As noted in Section 2, this approach has been widely adopted in similar studies. Therefore, in this work, we focused on applying both multinomial regression and decision trees to compare their performance and determine the most effective model for predicting academic trajectories.

Both multinomial regression and decision trees are well-suited for addressing multiclass classification problems, particularly when interpretability is essential. Specifically, multinomial regression offers valuable insights through its regression coefficients, which capture the direction and magnitude of each predictor's effect on the probability of belonging to a given class. By contrast, decision trees enhance interpretability by depicting the classification process as a sequence of decision rules that can be easily visualized and understood. These rule-based structures make the decision path especially intuitive for stakeholders such as educational authorities. Applying both methods to a multiclass problem not only strengthens the robustness of the analysis but also broadens the interpretability of the results from complementary perspectives.

Note that advanced post-hoc XAI techniques like SHAP were not employed, as the chosen models (multinomial regression and decision trees) are inherently interpretable, which was a core requirement of this study.

3.7. Survival analysis

Survival analysis is a statistical technique traditionally applied in the medical field, where the main focus is on the amount of time that passes before a specific event occurs. SA encompasses a range of statistical techniques that examine the duration between the start of a process and the occurrence of a particular event, aiming to identify the factors that influence the timing of that event [61].

Table 7
Coefficients and standard errors for multinomial model.

Coefficients				
	(Intercept)	Age	GPA	Credit Progress
B	56.8285	0.3917	-4.3396	-0.2798
C	58.9841	0.4888	-4.0745	-0.3528
Std. Errors				
B	1.1785	0.1633	0.6085	0.0524
C	1.1681	0.1721	0.6208	0.0533
AIC: 226.331				

The Cox regression model, introduced by Cox in 1972, is a semi-parametric model that combines both parametric and nonparametric elements. Commonly referred to as the Cox proportional hazards model, it assumes that the hazard ratio between any two individuals remains constant over time and is independent of the survival duration. Unlike the Kaplan–Meier method, Cox regression enables the inclusion of covariates or explanatory variables, aiming to model the relationship between these variables and individuals' survival times. It also facilitates the identification of factors that influence the occurrence of the event under study.

As with any regression model, it is necessary to estimate the coefficients (β). However, due to the model's semiparametric nature, traditional Maximum Likelihood Estimation cannot be used. Instead, Cox proposed the use of the Partial Likelihood method, which not only estimates the survival probabilities for individuals who experience the event but also accounts for censored individuals (those who have not yet experienced the event during the observation period).

The Cox model is given by (1).

$$h(t|x) = h_0(t) \cdot \exp(x'\beta) \quad (1)$$

Where:

$h(t|x)$ is the time (t) given a set of explanatory variables (x).

$h_0(t)$ is the baseline risk.

x' is the vector of the explanatory variables.

β is the vector of coefficients of the explanatory variables.

This analysis is well-suited to the study of school dropout because the phenomenon extends beyond a simple dichotomous outcome (whether a student drops out or completes their studies) and critically involves a temporal dimension.

4. Results

To generate the three models, the database was split 80–20, with 80% of the data used for training and 20% used for testing. Once the significant variables were identified and the assumptions verified, the Multinomial Regression model was developed. To perform this type of regression, it is essential to establish a reference category, assuming that the others can be predicted based on it. In this case, AT type A was established as the reference to predict AT type B and AT type C. Table 7 shows the multinomial logistic regression parameters, indicating that the variables with the greatest influence on academic performance are age, grade point average, and credit progress.

Therefore, in our Software Engineering program the predictions of categories B and C are obtained by Eqs. (2) and (3).

$$B = 56.828 + 0.3917Age - 0.339 GPA - 0.2798CP \quad (2)$$

$$C = 58.98 + 0.4886Age - 4.074 GPA - 0.3528CP \quad (3)$$

Once the coefficients of the multinomial regression model are obtained, a calculation is performed to determine the student's AT type, turning the task into a classification problem.

Table 8
Confusion matrix for Multinomial Model.

Observed	Predicted		
	A	B	C
A	20	3	1
B	2	10	0
C	1	0	48
Accuracy: 0.917			

Table 9
Multinomial model metrics results by class.

Metric	A	B	C
Precision	0.87	0.77	0.98
Recall	0.83	0.83	0.98
F1-score	0.85	0.80	0.98

Table 10
Validation methods for decision trees.

Method	Minimum precision	Maximum precision	Average precision
Leave-one-out cross-validation	0%	100%	83.43%
k-fold cross-validation	79.41%	86.76%	81.18%
Random Cross-Validation	77.64%	87.05%	81.88%

To predict the category, the probability of belonging to each class (type A, B, or C) is computed as shown in (4).

$$P(C_i) = \frac{e^{p_i}}{1 + e^{p_i}} \quad (4)$$

Where

$P(C_i)$ is the probability that the student belongs to the i th class. p_i is the prediction of the i th class given the multinomial regression models. The class with the highest probability is then assigned as the student's predicted category.

Table 8 shows the confusion matrix of the Multinomial Regression model, which correctly classifies 20 students who completed their degree on time (AT type A), 10 students who completed with some difficulty (AT type B), and 48 students who dropped out (AT type C). By summing the main diagonal and dividing it by the total number of data points, an accuracy of 91% is obtained.

Table 9 shows the values of the precision, recall and f1-score metrics from the confusion matrix in Table 8. Multinomial regression performs well especially in class C. Class B is the hardest to predict (lower precision), but still maintains a reasonably high recall.

As a second XAI technique, a model was generated using decision trees. Similarly to the multinomial regression model, the data were split 80–20. The decision tree was not pruned due to the small number of significant variables used, which implies a small number of nodes. To evaluate and validate the model, three validation methods were compared, as shown in Table 10. For this model, the best validation method was Leave-One-Out, achieving an average accuracy of 83%.

The decision tree generated to classify the three types of academic trajectories based on the entry profile variables is shown in Fig. 5. In Fig. 5 it can be observed that there is only one branch to classify students with a type A academic trajectory and one branch to classify type B. Table 11 shows the confusion matrix, where the tree classifies 69 students who graduated on time (type A) and 152 students who did not complete their degree (type C). The model achieved an accuracy of 80%.

Table 12 shows the values of the precision, recall and f1-score metrics from the confusion matrix in Table 11. In decision tree model, the class C has the best performance (precision and recall close to 90%)

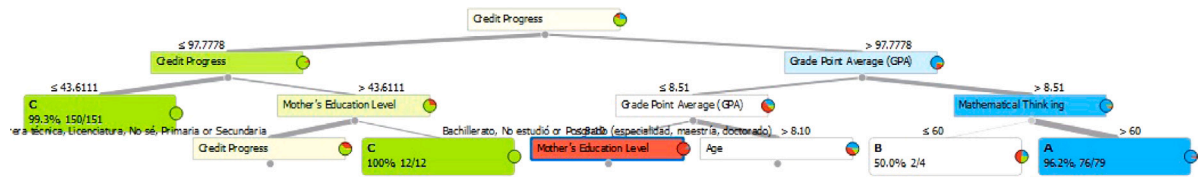


Fig. 5. Decision tree to categorize AT profiles.

Table 11
Confusion matrix for decision tree.

Observed	Predicted		
	A	B	C
A	69	10	1
B	16	13	11
C	6	12	152
Accuracy: 80			

Table 12
Decision tree model metrics results by class.

Metric	A	B	C
Precision	0.76	0.40	0.92
Recall	0.86	0.30	0.93
F1-score	0.81	0.34	0.92

Table 13
Rules to identify each academic trajectory.

For AT type A	For AT type B	For AT type C
GPA > 8.51	GPA ≤ 8.10	Credit Progress
Credit Progress > 97.77	Credit Progress > 97.77	≤ 43.61
Age ≤ 21	Age ≤ 21	

and class B is the most difficult to classify because it has low precision (40%) and low recall (30%).

To identify each AT, the model's characteristics were visualized, and rules were created. Table 13 shows the rules generated to identify each type of academic trajectory. For example, to determine whether a student will have a successful AT (type A), they must have a GPA greater than 8.51, a credit history of more than 97.77 credits, and be under 21 years of age.

As a third technique, the Cox Proportional Hazards model was generated. The database was split 80–20, with 80% used for training and 20% for testing. All qualitative variables were converted into factors, and survival time (the number of days from the student's start date to graduation or dropout) was calculated. The model was then developed using Backward Regression to obtain the significant variables.

According to Table 14, the significant variables for the Cox model were CENEVAL score, level of education attained, academic average, and credit progress. This model obtained an AIC of 621.73, which was the lowest among the models tested. Once the variables were identified, the Schoenfeld Residual Test was performed, which seeks to ensure that the variables meet the proportionality of hazards assumption under the hypotheses:

- H0: $p > 0.05$, the proportionality of hazards assumption is met.
- H1: $p < 0.05$, the proportionality of hazards assumption is not met.

According to Table 14, the significant variables for the Cox model were the CENEVAL score, level of education attained, academic average, and credit progress. This model obtained an AIC of 621.73, which was the lowest among the models tested. Once the variables were identified, the Schoenfeld Residual Test was performed to verify that the variables meet the assumption of proportionality of risks, under the following hypotheses:

Table 14
Coefficients for the Cox model.

	Coef	Exp (coef)	Se (coef)	z	P(> z)
Ceneval Score	−0.004	0.996	0.016	−0.245	0.806
Posgrad- uate	−0.457	0.634	0.240	−1.905	0.057
TSU	2.344	10.418	1.194	1.964	0.050 *
GPA	0.066	1.068	0.035	1.865	0.062
Credit Progress	−0.027	0.973	0.003	−10.03	0.000 ***
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1					
AIC: 621.7347					

Table 15
Schoenfeld test (DF = Degree freedom).

	Chisq	Df	p
Pcne	0.227	1	0.63
Promedio general	1.308	2	0.52
Avance crediticio	0.782	1	0.38
Nivel de estudios a alcanzar	1.853	1	0.17
GLOBAL	3.421	5	0.64

Table 16
Confusion matrix for survival analysis.

Observed	Predicted	
	No	Yes
No	33	10
Yes	1	41
Accuracy: 87.06%		

- H0: $p > 0.05$, the assumption of proportionality of risks is met.
- H1: $p < 0.05$, the assumption of proportionality of risks is not met.

As can be seen in Table 15, all the p -values for the variables are greater than the significance level (0.05), indicating that the variables in the model meet the assumption of proportionality of risks. The p -value for the general model (0.64) was also obtained, which also meets the assumption of proportionality of risks. This is visually represented in Fig. 6.

The model correctly classified 33 students who did not experience the event of interest, in this case those who graduated, and 41 students who did experience the event of interest, those who dropped out. The accuracy rate was 87%. Table 16 shows the confusion matrix for the SA.

Finally, Fig. 7 shows the ROC curve for evaluating the performance of the SA model. The area under the curve (AUC) is 0.97, which shows the model has a high ability to correctly distinguish between classes, making it a very reliable classifier for identifying dropouts.

To evaluate the predictive capability of the Cox model, two metrics were applied. First, the C-index (Harrell's concordance index), a goodness-of-fit measure used to assess the model's ability to discriminate between individuals, that is, how well the model distinguishes between individuals at lower and higher risk of experiencing the event of interest. In addition, confidence intervals were obtained using the bootstrap method to confirm that the generated model is both reliable and statistically significant.

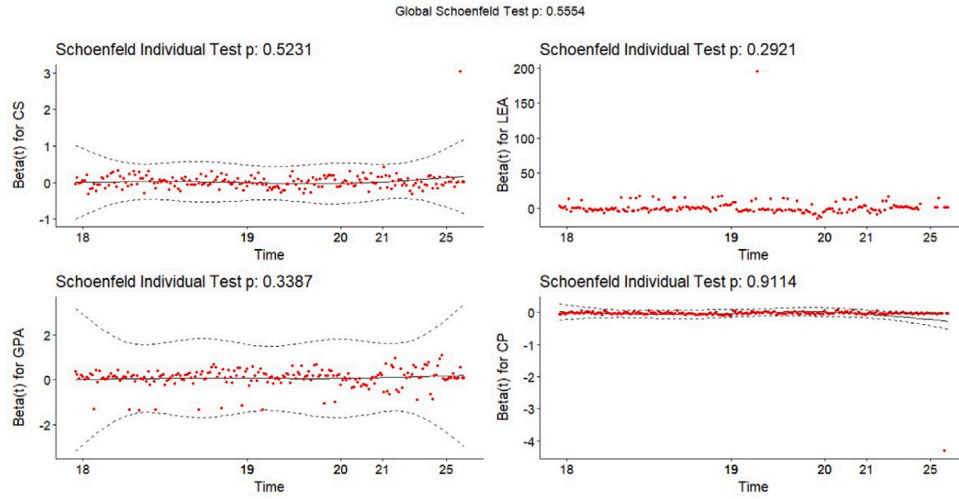


Fig. 6. Schoenfeld residuals.

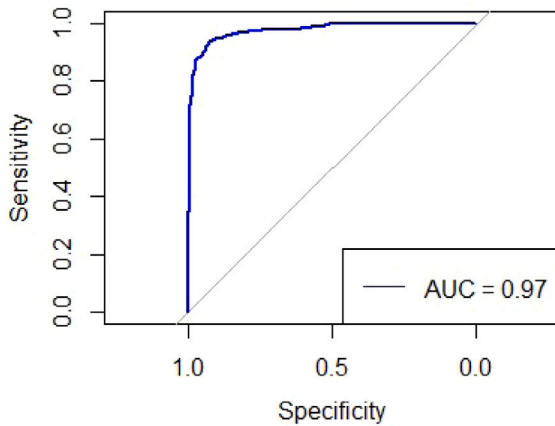


Fig. 7. ROC curve and AUC value for the SA model.

Harrell's C-index of 0.76 reflects strong discriminatory ability, indicating that the model can reliably distinguish students at risk of dropout. This level of performance is well-suited for identifying vulnerable cohorts and supporting targeted interventions, though further refinement is needed to enhance the precision of individual-level predictions. The 95% confidence interval ranged from 0.66 to 0.80, confirming that the model is both reliable and statistically significant, as the Harrell index value falls within the confidence interval.

As a second metric, the Brier score was calculated, which is used to evaluate the accuracy of the model's predictions. To compute the Brier score, three models were generated and compared: a reference model, which indicates the maximum possible error; a null model, which excludes significant variables; and the original Cox model. This comparison was carried out to verify that the proposed model demonstrates good predictive performance.

Table 17 shows the Integrated Brier Score IBS (at end point), where the Brier score is integrated over the interval $[0, t]$ described in (5), where $BS(u)$ is the Brier score at time u . Lower IBS values indicate better predictive capacity of the model. The results in Table 17 show that the IBS value of the Cox model is lower, indicating that the model is not only better, but also that the included covariates provide significant information for discrimination and prediction, so the model has predictive validity.

$$IBS[0, t] = \frac{1}{t} \int_0^t BS(u) du \quad (5)$$

Table 17
Integrated brier score ($[0; \text{time}=2345]$).

Reference model	Cox model	Null model
0.238	0.071	0.238

5. Threats to validity

Although we selected models that allow explainability through XAI to support decision-making on the most influential variables in student dropout or academic success, we identified some threats to the validity of our study. We next describe the threats to validity and how they were addressed.

1. **Dataset.** Although this dataset comprises students who began their studies between 2014 and 2019 (with those who enrolled in 2019 expected to complete their programs in 2024), it is acknowledged that the skills and knowledge associated with the admission profile may evolve over time. Accordingly, new cohorts of students will continue to be incorporated into the proposed models.
2. **Imbalanced class sizes.** While it is recognized that classification and survival analysis algorithms tend to favor the majority class, class-balancing techniques were deliberately not applied, as such procedures would artificially alter the natural dynamics of the dropout phenomenon. In other words, if dropout is indeed a recurrent event, it should be represented as a high probability of occurrence within the educational program rather than minimized through data manipulation.
3. **Generalization of models to other regions and profiles.** Although the data were collected from a single institution in Mexico and within a specific academic profile (Software Engineering), the potential for generalization to other profiles is acknowledged. Nevertheless, the main objective of this study was to identify the characteristics of students from this particular university. The approaches presented in this paper may also be adapted to other institutions, provided that the distinctive characteristics of their student populations are considered, as regional and disciplinary differences across undergraduate programs are to be expected.

6. Conclusions

This paper presents a characterization study of software engineering students based on data from their admission profiles and academic trajectories. Using statistical methods, supervised learning techniques, and Explainable Artificial Intelligence (XAI), it was found that there is a relationship between the admission profile and the academic trajectory of students. To classify this student population based on their AT, it is essential to focus on the following variables: age, CENEVAL score, GPA, credit progress, and the highest level of education the student intends to achieve.

At first glance, a high CENEVAL score might suggest that a student will graduate on time. However, some students with high scores did not graduate on time (AT type B) or did not complete their degrees (AT type C). In the case of AT type B, students may have experienced temporary withdrawals or delays in their curriculum. Conversely, for AT type C, the reasons may include unmet expectations of the program or insufficient academic performance.

Regarding the comparison of model accuracies, it was concluded that the best model for this classification task was Multinomial Regression, which achieved an accuracy of 90%, compared to Decision Trees and Survival Analysis, which showed approximately 10% lower accuracy. These findings will be highly valuable for educational authorities, enabling them to focus their efforts on monitoring these variables to support decision-making and reduce university dropout rates.

Beyond accuracy, multinomial regression offers superior interpretability compared to decision trees and survival analysis. First, the availability of regression coefficients allows researchers to identify which variables have the greatest influence on classification, whereas decision trees merely generate rules without quantifying the strength of predictors. Second, multinomial regression directly estimates the probability of belonging to each class, while decision trees rely on data partitions that often mix observations from different categories. Finally, multinomial regression enables classification across multiple outcome categories, in contrast to SA, which is typically restricted to dichotomous outcomes such as survival versus death (dropout versus culmination of AT).

The application of these approaches would allow educational authorities to:

- Identify students requiring early intervention, thereby providing timely support before the risk of dropout materializes.
- Implement tutoring and mentoring programs in collaboration with faculty members (teachers and students).
- Recognize students who are strong candidates for scholarships or financial aid opportunities.
- Revise and adapt the academic curriculum to more effectively align with students' needs and learning trajectories.
- Monitor non-academic factors that influence academic performance, including socioeconomic status, sociodemographic characteristics, employment conditions, psychological aspects, and other contextual variables.

Therefore, predicting dropout in SE students can generate significant social benefits, as it enables institutions to anticipate risk and provide timely, personalized interventions that improve retention, equity, and resource allocation. By making the factors that influence dropout transparent, XAI promotes accountability and empowers educational stakeholders to design fairer policies. However, these same tools may also introduce risks, such as predictions could stigmatize students, reinforce preexisting socioeconomic biases, or lead to overreliance on algorithmic outputs that overlook contextual or individual factors. Furthermore, results may vary across populations and institutions, raising concerns about fairness and generalizability. Therefore, while these techniques hold strong potential to support educational

improvement, their application must be accompanied by ethical safeguards to ensure they empower rather than disadvantage vulnerable groups.

Finally, this study advances the fields of Machine Learning and Computer Science by integrating explainability techniques with statistical approaches such as survival analysis and multinomial regression to address real-world, human-centered problems. Specifically, it examines students enrolled in the Bachelor's degree in Software Engineering at a Mexican university (a discipline requiring skills such as mathematics, logical reasoning, and certain social traits) demonstrating how advanced statistical methods and XAI can be effectively combined to analyze complex educational datasets. The proposed methodological pipeline not only enhances predictive accuracy but also provides transparent and interpretable model outcomes, thereby strengthening confidence in ML applications. Such integration reinforces Computer Science as a discipline that translates theoretical advances into practical, socially impactful solutions and highlights the applicability of XAI in critical domains including educational data mining, software engineering, and decision-support systems.

7. Future work

For future work, we propose the following:

1. To explore additional XAI techniques to verify whether the variable influences are consistent with those identified by decision trees and multinomial regression.
2. To increase the number of students in the study as more students graduate from the software engineering program.
3. To propose new variables of interest that may influence dropout rates, such as students' interest in the discipline, using data collection instruments or surveys.
4. To compare the results with data from other regions in Mexico to determine whether geographic location influences the classification of students with a Software Engineering profile.

CRedit authorship contribution statement

Isabel Vega-Rebolledo: Writing – original draft, Formal analysis, Data curation. **Angel J. Sánchez-García:** Writing – original draft, Software, Methodology, Investigation, Data curation, Conceptualization. **José Juan Muñoz León:** Supervision, Methodology, Data curation, Conceptualization. **Jorge Octavio Ocharán-Hernández:** Writing – review & editing, Writing – original draft, Software, Conceptualization. **Karen Cortés-Verdín:** Writing – review & editing, Validation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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